

Xiongxin Yang

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🏠 yangxx0405.github.io

Research Interests

Discrete probability; Markov chain Monte Carlo method, Markov random field; statistical and learning theory; computational and discrete geometry; mathematics for AI, AI for mathematics.

Education

University of California, Santa Barbara

Sept 2025 – Present

Ph.D. Student in Computer Science

◦ Advisor: Prof. Eric Vigoda

Northeast Normal University

Sept 2020 – Jun 2024

B.S. in Computer Science and Technology

◦ Advisor: Prof. Zhiguo Fu

Employment

The University of Hong Kong

May 2025 – Jul 2025

Research Assistant hosted by Prof. Weiming Feng

New York University Shanghai

Oct 2024 – Jan 2025

Research Associate hosted by Prof. Jie Xue

Visits and Internships

The University of Hong Kong

Mar 2025

Visitor hosted by Prof. Weiming Feng

Nanjing University

Apr 2023

Visiting Student hosted by Prof. Yitong Yin

Selected Awards and Scholarships

Presidential Scholarship (Northeast Normal University)

2022 & 2024

Ranked 1st in the major

National Scholarship

2021

Top national scholarship in mainland China (top 0.2% nationwide)

Publications and Preprints

Unless noted otherwise, authors are listed alphabetically, following standard practice in theoretical computer science. ● denotes conference publications; ● denotes journal publications.

[8] Exponential-Time Approximation Schemes for Ordering Problem

Huairui Chu, Ajaykrishnan E S, Viktoria Korchemna, Daniel Lokshtanov, Anikait Mundhra, Saket Saurabh, Thomas Schiber, Xiaoyang Xu, Jie Xue, **X. Yang**.

In progress

[7] Simple and Efficient Algorithms for Approximating Maximum Depth of Unit Disks and Unit Balls

Sujoy Bhore, Jie Xue, **X. Yang**, Jiumu Zhu

Submitted

- [6] Fully Dynamic Diameter Approximation in Unit Disk Graphs
 Sujoy Bhore, Jie Xue, **X. Yang**, Jiumu Zhu
 Submitted
- [5] Near-Optimal Dynamic Data Structures for Maximum Depth and Klee's Measure of Boxes
 Sujoy Bhore, Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
 53rd EATCS International Colloquium on Automata, Languages, and Programming (**ICALP 2026**)
- [4] Learning CNF formulas from uniform random solutions in the local lemma regime
 Weiming Feng, **X. Yang**, Yixiao Yu, Yiyao Zhang
 58th Annual ACM Symposium on Theory of Computing (**STOC 2026**)
- [3] Dynamic maximum depth of geometric objects
 Subhash Suri, Jie Xue, **X. Yang**, Jiumu Zhu
 41st International Symposium on Computational Geometry (**SoCG 2025**)
 - Invited to SoCG special issue (regretfully declined)
- [2] Spectral Independence Beyond Total Influence on Trees and Related Graphs
 Xiaoyu Chen, **X. Yang**, Yitong Yin, Xinyuan Zhang
 2025 ACM-SIAM Symposium on Discrete Algorithms (**SODA 2025**)
- [1] Beyond windability: Approximability of the four-vertex model
 Tianyu Liu, **X. Yang**
Theoretical Computer Science, Volume 995 (2024), Article 114491

Talks

- Dynamic Maximum Depth of Geometric Objects
 - The 41st International Symposium on Computational Geometry (SoCG 2025)
 Jun 25, 2025 @ Kanazawa, Ishikawa Prefecture, Japan
 - The 16th Annual Meeting of Asian Association for Algorithms and Computation (AAAC 2025)
 May 31, 2025 @ The City University of Hong Kong, Hong Kong SAR, China
- A Glimpse into the Theory World
 - Information Science Academic Forum (in Chinese)
 Apr 11, 2025 @ Northeast Normal University, Changchun, China
- Approximability of the Four-Vertex Model
 - NYUSH Theory Lunch
 Oct 21, 2024 @ New York University Shanghai, Shanghai, China

Grants

Catalyst Grant Program (Cohere Labs)
 \$1000 API credit

2026

Academic Service

Conference Reviewing: SoCG (2025), WWW (2026)